

Marilyn Stendera ~ Active Inference Insights 005 ~ Autopoiesis, Heidegger, Futurity

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hello everyone and welcome back to active inference insights I'm your host Daris PV Wayne and today I'm talking with Marilyn Stendera Marilyn is a lecturer in philosophy at the University of Western Australia her research focuses mainly on the phenomenological tradition particularly its intersections with cognitive science and she has a particular interest in the role of time and how it shapes our cognition and experience in 2015 she wrote the paper being in the world temporality and autopoiesis which explored the inextricable intertwining of age in the world the Confluence of indigenous perspectives and in activism and self-organization and to my eyes at least a focus on these domains makes her work amenable at least in principle to the insights of active inference and among other things it is the viability of that project that sort of coming together that we will be investigating today Marilyn thank you so much for joining me and welcome to the show thank you so much for having me Daris I'm really pleased to be here yeah yeah it's an absolute pleasure I mean um I sort of came across you by finding that 2015 paper and I love it um I went back and read it again uh sort of in preparation for our podcast today and it's it's so insightful um and yeah I I just as a starting point for anyone listening I highly recommend they go check that out um but let's dive into some of the things you spoke about in that paper there's a quote that you took uh that you referenced from Varela um Francisco Varela who who sort of introduced the term autopoiesis with Maturana into the biological literature in 1980 and he wrote a universe comes into being when a space is severed in two I wondered if you could help us unpick uh what he meant by that yeah so that's uh that's um so far thank you so much for your your kind words but Myer I think everybody loves hearing that so um yeah so Varela in general has been a huge uh influence on how I look at cognition I think since I wrote that paper I've kind of come to um more newton say yes understanding of what outer PES and autonomy and activism are about but yeah this this this idea of um of the cognitive process as being fundamentally a relation and precarious and uh something that can't be understood by looking

at at any of the one components involved in that process in isolation right it's that fundamental idea of of in a sense cognition is is always this precarious process in attention um it's the creation of a boundary it's the creation of a separation a barrier but also always at the same time uh a barrier that has to be porous and a barrier that that uh is always something that is under threat that has to be maintained um and there's there's always that question which is something that's fascinated me of uh the the the kind of the tension between just how much that barrier has to be maintained versus how much it has to be open to change which is I think also one of the the really interesting points of of conversation with with inference but yeah this this idea of of of cognition World and Action there is there is a boundary there's a separation there is this this sense of um what in subsequent kind of inactive literature is called uh both self- production and self-d distinction especially right so that something brings itself into being something maintains itself something distinguishes itself and thereby we get this kind of whole complicated wonderful condition of having world right fantastic gay it's interesting that you mentioned that there's a kind of philosophical question there about the point to which uh you know where does where does that self- distinction uh start and end um yeah I mean I think a lot of the active influence literature is converging on a a point of vagueness sort of a soriz paradox perspective that you know when does a grain of sand become a pile or a heap and and so on um I guess another philosophical question is does it you just to start with a really fundamentals here about like this coupling does it even make sense to think of anything as having a kind of autonomous uh essence or nature without some coupling environment that's a brilliant question and so ultimately in a sense I think what a lot of the the literature that my work moves and wants to do is move away from this idea of having an Essence right so there's there's already that kind of Prior question of can you have an entity can you have a system can you have an organism that that is clearly kind of itself and is clearly kind of distinctive without having you know anything as as kind of substantive as as an Essence um and what does it mean can there be a kind of I guess non-teleological concept of of what it means to be alive that so some folks look at it or you know yeah just is is there a way of talking about Essence without talking about Essence um in a sense um and so yeah so so overall to me I think and I think all of the discourses that my work moves in and and what strikes me about active inference as well is this idea that yeah coupling is is fundamental there is no such thing as a non-relational at least not a non-relational living system a non-relational cognitive system cognition life is always a matter of structural coupling um and yeah that's that in a sense is kind of what we have to start from sure yeah I think in I think that's a good point that uh we're

definitely skeptical of substant substantive Essences sub substantive say it right that time uh selves in the sense that active inference casts uh thingness as a statistical distribution um and sort of uh builds off dynamical systems theory about this kind of idea of attractor sets or sort of the peak of probability distributions I actually just reading Manuel delanda's book that came out last year the phenomenology of the materialist phenomenology and he without even referencing active inference talks also about these kinds of basins these attractor basins and I guess if we did want to just take a broad rush I guess we could start defining things as being in some ways the statistical manifold or or or this attractor base and that's cool that's yeah that's a nice perspective um so let's let's sort of Jump Then to autopoesis um maybe it'll be useful for you to just give our audience a kind of overview view of what that term means um sort of coming from the verela literature and then we can kind of fold in active inference yeah awesome that sounds great it's it's funny that you mentioned that book I literally just bought that last week it's flying around te it's good it's good I'm enjoying it yeah awesome that's that's good to hear I look forward to reading it yes so I'm so I I should flag firstly that you know this this is like my um attempt to explain what these concepts mean so of course on the one hand there are very kind of clear definitions especially in a very landmar onest initial proposal of of these terms but since then the whole kind of you know so-called Ultra poetic and active approach a term that in itself is is contested because lots of people point out it's not just about aesis anymore um all of these terms are are you know super contested and um their whole literature is about you know kind of defining them in relation to each other and so yeah so I umum I kind of I I follow a bunch of of contemporary folks in and cashing out um outter pois as as a kind of a a type of organization um wherein an organism essentially uh both produces and reproduces itself and distinguishes itself um and so uh one of those contemporary folks is Ezekiel deer who conveniently has those those self- production self-d distinction and he points out you know you you can't have an out system without having both of those so you can have a system and that kind of manages to Define and distinguish itself but if it doesn't reproduce the kind of processes that um that Define it then it's it's not autopoetic and you can have a system that is able to kind of reproduce its components and processes but if it's not able to maintain kind of a precarious boundary um then it's it's not autopoetic either and and so that this is the idea that this is this is kind of an an Oran organizational principle um so in in verand matan's original account this was meant to be a definition of life as such essentially um and since then it's become a lot more complicated so of course ver and Matana themselves developed this um much more extensively and these days I guess the more appropriate term to

kind of use this autonomy right so there's this kind of general idea of a self-producing self identity that can be kind of instantiated in lots of different ways a lots of different um structures and uh there's there have kind of been lots of other tools added to the to that conceptual repertoire so one of the reasons uh so one of the the kind of the main Innovations for example has been the addition of this this concept of uh of of adaptivity right so where outo pois is no longer seen as as kind of being sufficient for capturing what cognition is all about there's also this this idea that every living system every cognitive system has essentially produces these kind of Standards on all do these things how do these these kind of fundamental features both scale down and scale up and so that idea of of aiming to do well rather than aiming to do worse so there is there is the IDE here and I do think it's it's plausible that this is something that happens you know at the most basic level of of life and cognition that's kind of the where the switch flips uh in a sense to to a living cognitive system um and and you notice so for you know for for this kind of branch of an activism life and cognition are essentially intertwined so've got a living system you've got a cognitive system and so I think um I think just the the capacity to respond to uh to to stimulate and and circumstances and what's happening in the environment the capacity to kind of have a meaningful perspective um in kind of not a very you know thick psychological sense but just in kind of basic respond to the Seline that way rather than that way I think it is you know it is it is absolutely plausible that that's kind of a fundamental aspect of what it means to be alive and so then the interesting question kind of is how do you get from that how do you get from you know a bacteria moving away from areas where there's I don't know less nutritional concentration or where it's bad stuff moving towards the good stuff how do you get from to kind of our complex reflective metacognitive psychological processes and yeah I think that's that's kind of where a lot of these discourses can kind of come together to flesh out that developmental scale that scale of cognitive complexity how does this basic responsiveness this basic kind of world involving World producing not wanting to die aspect of life how does that then scale up to the very kind of complex reflection that we're able to do and it's yeah it's really interesting I think to consider how active inference can kind of potentially contribute to that maybe quantify some stuff add some tools to the repertoire yeah I think again I can't speak for everyone in the field uh because I'm like everyone else I guess kind of a newcomer um well at least for a lot of people but I think what's really fundamental to the complexity of cognition under an act of inference framework is the depth of temporal planning so again as I said in my introduction you're really interested in time and I think what what we're finding is that like a complex

organism like us what might make us definitively complex as our capacity to do counterfactual reasoning so take your example of a bacterium um it it can't it doesn't have a model of the world where it can sort of at least not in any significant way use um retrospective inference to inform future inference so it doesn't know for example okay there's neutri over here but actually I might suffer that long a short-term loss for the long-term gain of being in this region right so it exploitation uh exploration distinction is like it's just less nuanced I guess um they're humans how does that sort of sound to your ears that maybe what we're looking at here is a uh depth of temporal planning yeah that that sounds great so um generally speaking so um I I love the idea and this is something that I kind of try to pursue um of cashing out the spectrum of of cognitive complexity in terms of temporal complexity right so largely you know including along the lines that that you've just mentioned and so there's there is some you know really interesting literature already on this that kind of points out um some of the the ways in which um yeah those those basic structures of kind of future directedness and and anticipation and planning can become more complex especially through as you said right through our relationship to essentially more complicated kind of future right so counterfactuals unpredictability uncertainty uh there's there's some cool new work looking at you know spikes of entropy um that sort of thing and that to me is also one of the the places where really the phenomenological tradition becomes super relevant and and super interesting right because there's there's this kind of consistent IDE IDE that um as and this this this starts before in a sense so not quite before but a little bit outside of phenomenology in the work of Al bur and this idea that uh so for him cognitive complexity and complexity of embodiment is explicitly linked to kind of more radical confrontations with uncertainty which also comes with those confrontations being more meaningful right so the more complicated you get the difference between us failing and a bacter and failing there are all these existential meanings risk failure uh carries so much so much more significance in in the scaling up of those mechanisms and then you get that in the phenomenological tradition too this this idea that um yeah we we have more and more complicated Futures but also you know kind of more complicated relationships between our own histories and the future and um and yeah I I like the idea of of as you said you know cashing that out terms of of more complicated different richer forms of things like anticipation yeah there's so much there um I mean yeah I've been saying stuff that you know people are even grounding Consciousness or or you know trying to resolve the hard problem through this route so the the you know maybe the function of Consciousness let's not get to the kind of deep qual I don't think we need to go there is to kind of you it's it's fundamentally

episodic to use what's happened in the past to inform what's happening uh you know you know what's going to happen in the future or what might happen in the future um yeah that's that's really beautiful and that has a as you say kind of scale up to significance I mean it made me think of Ern Becka and the denial of death um and this idea that like that finality I mean again and Hyer that that finality is kind of fundamentally framing of our being in the world um as you can tell I you know I like to use this podcast also to point people in directions um of where they could read to uh or or listen to to learn more about these things I think a really wonderful paper um outlining some of these themes is by Andrew cor Coran javanni Petula and yob Hoy um from our static agents to counterfactual cognizers that's a 2020 paper and it's really wonderful um so anyone who's listening who's interested in maybe like how are we going to carve out these distinctions between simple organisms or non-active organisms simple simple living organisms complex living organisms they do a really good job giving a kind of framework in which to to assess those those ideas um it's another sort of interesting philosophical question that's really rearing its head at the moment for some reason there's this distinction between realism and instrumentalism um I was wondering you know that it's a Live question in active inference for example whether the the mark of blanket which acts as the boundary between internal States and external States whether that's a instrumental tool for us as scientists or whe that actually had some kind of true ontological status I was wondering if you'd had any thoughts about whether autopoesis so self-organization uh self production and self distinction is a uh just actually an instrumental an instrumentalist thing so it's just a way of us to describe what seems to be happening um at different levels of biology but it can even go broader than that or whether it's actually an imperative it's actually you know a genuine thing it's a mechanism that's occurring at it within these organisms yeah that's a that's a really really interesting question darus thanks for that and yeah I um I did see when I was I was trying to do a little bit of homework um for for this and then trying to read up on an active inference a bit I I saw there there was one paper that kind of made a throwaway remark of you know whatever the metaphysical stages of the mark of blanket might be and I thought right what's that about so that I think that that kind of explains explains what that was but yeah I mean it's it's in a sense that's kind of a a fundamental metaphilosophical and methodological question in again kind of all of these discourses right it's it's it's something that affects all of these different models of cognition something that you find in the phenomenological tradition too like are these these structures are they heuristics are they kind of you know the best possible thing that we can come up with to to make sense of what's going on or are they actually out there what

does it mean for anything to actually be out there in this sense and on the one hand I I don't want to you know get away from the kind of the materiality of cognition which is you know kind of crucial to all this like you can see this stuff happening um you know in different kind of cognizers and I guess it becomes these are structures that become more and more heuristic and instrumental as you kind of move up in cognitive complexity and you know when it comes to especially kind of structures like outer PES and autonomy I mean you know get applications of this sort of thing to the behaviors of groups for example and you know you get this kind of scaled up in things like systems theory and I guess there it becomes more and more um instrumental but one thing that kind of comes to mind uh as you're raising that point is this kind of this there's really interesting insistence in the corner of an active discourse that interests me the most in the embodied mind that there is we don't have to and this is kind of deeply phenomenological as well we don't have to worry that much about these questions because there is this kind of we do have an insight into what is really out there so to speak and into the processes of life and the processes of cognition because we're living cogniz as ourselves there's this idea which kind of actually Echo I think you know hiig his famous idea that like we're particularly well positioned to do ontology because you know where these being who have insight into and who care about the structure of being itself and within yeah in you know the embodied mind there's this idea that as living cognizers being alive and being cognizers the same thing gives us an insight into the structures of life and cognition so kind of any heuristics that we use even if they do end up being purely instrumental heuristics kind of give us something genuinely important and insightful about what's happening out there because in a sense for us there is no such thing as you know just a neutral out there sure there kind of interesting idea of like a privileged Insight um into cognition by virtue of of being cognizer so I guess yeah as in terms of the phenomenological tradition too it's less of a question that I'm particularly concerned with it's interesting but yeah there's there's there's more of a sense of you know let's look at what this stuff is like from the inside without being too concerned about what what kind of actually is out there so to speak fair enough um I guess like the the harder scientists might you know their ears might prick up when they hear that actually our stance the fact that we take a stance on the world uh yields a greater possibility for insight because they might say well every insight you're going to have is going to be colored in a way by your own personal experience your history your cultural baggage I mean very postmodern take as well so do you also in a sense do phenomenologists have to recognize the bias or yes the coloring of these insights and to

what extent therefore is objectivity of the cards for a phenomenologist yeah that's a that's another great question and yeah absolutely so in a sense the idea that you are that any insight that you come up with any any knowledge any claims are always going to be situated um they're always going to be richly shaped and informed by not just your personal history but of course you know broader history social norms um your you know kind of cultural expectations um the situatedness of if knowledge is kind of a fundamental starting point for phenomenology and so it's less about taking objectivity of the table and more about redefining what objectivity is and so I think that's that's one of the the kind of key points of resonance for example between phenomenology and and an activism right so it's is this idea that again knowledge is always fundamentally situated it's always historical it's always uh socially culturally emo uh embodied and embedded um and so that means we have to think about um what it is that that we're actually aiming for or and and you know there there's there's no problem in getting rid of this this or in not being able to achieve this kind of sense of of neutrality that isn't really possible for a sense making organism anyway right because we enact our world um and we we bring forth our worlds in ways that mean we're not going to encounter uh the world in in kind of neutral you know gray pure brute matter terms um and yeah that's that's I mean that's you know a widespread point across a whole bunch of discourses the interesting question then is how do we reconfigure what we actually mean by objectivity and especially of course how do we build into subjectivity into that um and yeah so I think you know that's that's the idea that knowledge is is fundamentally situated is is is very much it's not a problem for phenomenologists of course there is you know there's this this kind of history of well phenomenologies trying to discover these you know General structures that make cognition possible um so isn't there you know some kind of tension in saying all knowledge is situated but also here are the universal structures that make all situate knowledge possible so that's kind of that's a that's an ongoing kind of interesting tension and problem but yeah um I think it's it's more about redefining what kind of what we mean and want by objectivity well it's a you know it's a uh self defeating element of kind of any nonobjective or or theory that postulates non-objectivity right I mean like if the postmodern claim that truth is relative is relative then that claim itself is relative so anyway we it's a it's a kind of Wormhole that we don't necessarily need to go into I I mean that it would be also interesting to have a mathematician on here um because I could imagine they would have some some gripes with our inability to get hold of the object you know the objective but again there's no need to go to go there um I did have a question though um I mean again there might

be one of those that are isn't too relevant to the discourses that you're engaging in but it's a question sort of about ends um and and purposes so you write in um one of your papers I'm not sure it's the 2015 one it might be another one that an autopoietic system is autonomous in that it is its own product it does not exist in order to bring about something else to accomplish some externally defined end but rather to be and perpetuate itself its identity is its task you also say in the same paper the organism's purposive aiming towards the overall end of survival as well as its more immediate needs in successfully navigating environmental circumstances are structured by a directedness towards something which has not been reached yet I guess at first glance there might be a superficial tension there um can it be dissolved just by saying that what we're talking about in with perpetuation um in the first quote is survival in the second quote is that to do with the kind of identity conflation is that the way out of that potentially tricky tension yeah yeah so that's uh that's more just kind of a a problem of um the that particular expression so the the point about um an organism's kind of identity being its own task rather than anything else it's not to say that you know a cognizer literally does doesn't have anything else to do except VI um it's it's more about this idea that in order to count as an as you know an autonomous living system um it has to have endogenous ends right and so it's survival has to it's it's kind of its its drive to to you know persist to maintain itself to resist the possibility of annihilation has to come from within itself it's not something that you know it's kind of one of the classic ways of distinguishing say an an artificial from an a living system or a living from a non- living system right it's it's not a living system is not something that has been given its Ends by somebody outside itself like say a designer or a programmer uh and where its end is you know to do something whether it's it's you know a line of code that's meant to do something or whether it's I don't know a sort of basic machine intended to do something the point of the living organism is itself is its maintenance of itself and then of course in pursuit of that self-maintenance that resistance against you know the constant threat of annihilation it pursues all sorts of other projects which you know you can kind of think of if almost is and I think this is why Heidegger is especially helpful it's kind of arranged around you know this this kind of fundamental self-concern and you know as organisms get more complex you get more and more sort of complicated project arranged around uh what Heidegger conveniently calls you know the for the sake of myself um thanks H and yeah yeah no gotta gotta love the Heidegger but um yeah yeah and you know and there is there's those projects kind of take on more existential uh significations as well and what it actually means to be self-concerned is no longer just I want to not starve but you know I have a particular conception of

what I want the story of my life to look like sure yeah this is a really interesting Avenue because it sort of rears its head in active inference literature as well active inference conversations because I think it's still a Live question I might be wrong but I see it still as a Live question is to what degree is all the organisms do to what degree is that just a minimization of free energy and I guess the problem at least in active inference is that basically everything can be cast as B optimal so this is called the complete class theorem so even if I do something which seems antithetical to my survival under some priors it's Bas optimal and so I guess we either could cast this as not being a form of self organization or we could just expand the notion of self I guess in a deflationary way to mean everything that a thing does and it's just based optimal under that scheme so again it might be a kind of um empty question but just to say it's it's live at least um within Act of inference as well I mean you know you thought about this on the inactivist or autopoietic end it was a question I had which is you know I think you you you said both in your paper and here that aesis is certainly necessary in the same way that minimizing free energy is necessary in some quite trivial sense for the Persistence of an organism but it might not be sufficient to Define exactly what that organism is again that's the open question um if it's not sufficient just it might be a very silly question what else do organisms or cognizers do yeah absolutely see so generally speaking um autois and autonomy aren't you know they they are seen as necessary but not sufficient conditions and so that's where uh the concept of adaptivity which I touched on earlier comes in right so kind of one of the the kind of key discourses uh has for the past probably couple of decades or so maybe a bit less actually maybe closer to like 15 years suggested that yeah that the necessary and sufficient conditions for cognition and life are autonomy and adaptivity right so that that idea of being able to respond to whether you're doing better or worse and and steering away from from doing worse so that the idea is that that might be a kind of you know basic enough condition that can still be instantiated by lots of very very simple organisms but that can scale up in all sorts of interesting ways including um you know some of the the cards that we talked about but there is you know there there is still kind of this this live debate and this um this this ongoing question of yeah is is that enough in terms of sufficient conditions for life and cognition but also generally you know um how do you draw the line between uh is that enough to draw the line between a living and and a non-living system and I know that um so like I said I I you know I tried to do a little bit of of reading especially on on some of the dialogues between folks working in active inference and folks working in an activism before this and I saw you know that's kind of been one of the questions where folks have kind of

asked each other you know maybe maybe this is this is something where these perspectives can collaborate it seems like you know across all of these discourses there's the concern it looks at these theoretical Frameworks can apply to all sorts of different systems how do we for example go from this to explaining what it means for an organism a living organism to have a meaningful perspective do we have enough resources to be able to flesh out what that is and how it becomes so much more complicated um and yeah so I think I think that kind of that continues to to remain a live issue in lots of really interesting ways sure and also quite a at times heat debate uh between and researchers which I will not get into uh I want C out to that um let's let's go to haiger then uh because he's a sort of uh specialty I guess of yours or or focus a topic of focus um you know scanning through just the active inference literature researchers might not know too much about haiger there have been papers on hle Merlo Ponte obviously is very important because of the embodied aspect of his work um I inip poito on last week and she spoke about Simone DEA and how her existentialism can be married to act of inference but hiiga for some reason seems to be uh the elephant in the r room who never really gets mentioned I think a good place to start and you're going to be far better place to do this than me is to tell us why Hyder go is just so radical uh and my thinking about this is just framing him as kind of the anti- day cart so sort of setting up that philosophical history that might be a complete broad brush uh misidentification but where the kind of cartisian tradition was uh and how that kind of got collapsed by haiger and others might be a good place to start yeah absolutely and and so so one thing I I I'd want to say at the start too is that so even within an active discourse there hasn't been that much um looking at at hiig go broadly speaking so it's always you know as you said Merlo Pony hor it's great to hear that um Min is is doing some work on some Ro run because yeah I um I've done some work on her myself and from what I know of her I I think I'd agree with um with that perspective so yes yeah it's it's interesting that there is this this kind of significant Gap and so in terms of what makes hiiga radical and and interesting and useful um in kind of opposing that cartisian tradition so uh to me the first thing that kind of stands out and this is something that I touch on in that that older paper that you know that um that you came across being in the world um temporality out poesis is that the way he talks about the relationship between cognizer and world so I'm using the word cognizer I don't read heiger in a particularly Orthodox way I don't particularly care about reading haiger in an orthodox way either um and so yeah so the the way that he talks about the cognizer world relation is both startlingly similar to and also particularly useful for uh all of these kinds of discourses that look at cognition as active situated um action oriented

being about you know meaning generation being of relational right because in in haig's account of being in the world you essentially have the same problem that we're we're still talking about in these discourses which is on the one hand emphasizing the kind of the relationality the the extent to which cognizer and world are intertwined such that you can't understand either part from the other why while on the other hand not kind of completely dissolving the cognizer into the world right so unlike you know some models of say for uh of you know embodied cognition there is still this sense of holding on to a core of self-concern um and this is this is why I argue it's especially resonant with the inactive tradition holding on to this this idea of those boundaries right we've again we've got that central question of uh a living system is one that defines itself it has to be a poorest boundary but it nonetheless has to be a boundary so again for hiding of the subjects not dissolved into the world it's fundamentally it produces the world it enacts the world it's integrated with the world and yet it still has that kind of core of of self-concern and and of of what you know an activist call adaptivity this kind of future directed self-concerned planning and so um yeah as phenomenologists are often accused of kind of being unfair to De card and and I do think in many ways we are but you know as of thinking as of the cartesian tradition more specifically than de there is yeah that that kind of radical insistence on undermining dualism insistence on undermining separation of mind and World subject and object but also mind and brain brain and World subject and and body right this idea that these are all kind of fundamentally constitutively uh connected um and so I know this has already been a long answer one other thing I I wanted to throw into and this is something that I think is especially underappreciated in all of these literatures is how hiig then links all of that to time right so all of the phenomenologists um and this is something I I try to argue too like time is fundamental to phenomenology every phenomenologist has an account of time uh and you know and they're all cool and great and interesting um hiig accounts stands out in particular because it is fundamentally linked to practice it's fundamentally linked to kind of salient responsiveness to self-maintenance to motivation to this kind of very basic practical normativity and to what we've been talking about that you know that kind of temporal complexity that that you know that sense of of like life itself is fundamentally kind of anticipatory and that then gets complicated in all sorts of rich and cool ways and so so yeah so you've got this model of time that's kind of fundamentally linked to practice normativity motivation self-c concern adaptivity so yeah my my contention there is like hey look if if you know if we're looking at time and and time is super important to all these embodied cognition discourses why not look

at this model of time that kind of comes prepackaged with this idea of of relational cognition and practice sure yeah yeah I mean the parallels there or at least the the way that Ina activism and active inference are clear well even if it's implicit clearly informed by hiigaran perspectives is is kind of very transparent they're very obvious um yes I think the disdain that later phenomenologists have for dayart might be influenced by the disdain that the earlier phenomenal had just had for dayar I think there's a good I was researching up I just think there's a good story about um Hider throwing sra's book in the bin because SRA sort of reinvoke Cartesian self you know probably quite an angry chap as well um Tiger's personal life um perhaps there is there is the not egg there yeah there's a lot that I mean actually the on complete side note there's um the the history of philosophical arguments between like you know supposedly quite like stoic uh Placid men because it's all men is is a is a fascinating I read a really good book ear this year about um the this fight that Vicken Stein and K POA had at cambridge's philosophy Society um which I recommend highly it's called Vicken Stein's poker anyway it's a complete side uh side note um I think it might be quite useful for people to because they might not know what you mean by practice uh and this notion of practice and Tool and and how this all links into being and being in the world perhaps an example would be useful because I think I've been thinking about this because I I just wrote a paper with some really wonderful co-authors as well so credit to them um about Flo and my thinking about Flo States is that it's it again this is a Live question but it might be rooted in the same mechanisms um that draus called skillful coping and hiding word of called a-h handedness right so maybe rock climbing is kind of part of the same category as opening a door or gripping a water bottle right um so I think you know this might again this is another Live question is the degree to which uh people conceive themselves as cartisian egos there might be an intuition at least when we're talking intellectually to say well I am this rified self that has this kind of Separation in time and space and identity from the things on which I act perhaps you can give us an example just from everyday life where hiiga reveals to us that that Intuition or that uh perspective might not actually correspond to the uh the phenomenology yeah that's a that's a great point and um just on the the the wienstein book I think I came across the same one it also talks about fantastic case where I think he he threw chalk or keys or something at one of his students head and it actually yeah I think he was like seemly quite abusive um yeah but also yeah obviously extremely accute and intelligent I'm not going to demean VI canstein I again these also not here to defend themselves so who am I to who am I exactly there is there is there is a point to this right so there is kind of I think you can always see this this like this this

continuing of in a sense a cartesian tradition of thinking about what it is that philosophers do and the kind of activities that are important where it's there this idea that you can divide what's you know what's happening in here the really important stuff from you know kind of acting in the world and then practicing in the world um and that you know yeah these these so-called Geniuses they get away with terrible behaviors like that because we we privilege we prioritize what's going on in KS as opposed to to kind of how they actually act in in the world and so yeah so coming back to um your point then about um about practice or just in terms of what I mean when I talk about practice it's just a kind of the very basic means ends kind of connection that you have Through Time right and so controversially and and I don't think other hog garant would agree here it's you know even like uh bacterium swimming away and and swimming towards you know a better nutrient gradient is is kind of like a primordial instantiation of practice and also get you know more complicated as you said with with how equipment and Tool being so just to to kind of to give a quick overview of those Concepts and then maybe try and come up with an example um so uh for hiig together it's it's something that will be familiar to to folks who work in you know embodied cognition or potentially even active inference it's this kind of overturning of the traditional hierarchy of how we do things where the idea is that you encounter the world in Practical terms first before you then kind of encounter them in you know more Quantified theoretical terms so that you know you're not and and I yeah I think even within active inference this holds like you're you're not you know the Terminator walking around with kind of you know numbers scrolling across your vision figuring out you know that object classified as a table made out of cedar wood is you know three met small so on and so forth you you encounter it first in terms of well I've got heavy stuff I want to put stuff down um or you know I don't want to bump into the sharp edge of that table and you're not even you know explicitly thinking about this necessarily in such terms it's more about encountering the world in terms of calls to action uh what folks in ecological psychological affordances you know different possibilities for for doing things um and there is I think a risk of of kind of under under inter Eliz this process right so fer there is still very much this sense that what you encounter isn't isn't like you know this this kind of just pure mass of complete non-conscious absorption you're still kind of encountering things as things and there is still very much the possibility of at least transforming some of those practical relationships into for example linguistic relationships if you need to um but yes so in terms of examples I mean you know heiger gives us the classic example of the hammer yeah um but something that I actually like using uh in class when I talk about this uh is so I I I game a lot

too much um and I feel like this is something that fundamentally kind of made hiding as account of practice more feasible to me right because you know say you're you're playing with keyboard and mouse or you're playing with a controller um you are very much so immersed in the game world world that you're not really consciously reflecting on uh these these tools that you're using you're not you know you don't have this kind of consistent uh metacognitive reflection on yourself as being separate to the mouse separate to the control you're just you are just acting right um and it's it's when they as you know Hada famously says with the H it's when they break down that yeah they become apparent as objects so that the kind of the terminologists like do sorry they're like using so-called breakdown cases because the idea is that we're so immersed in kind of everyday practical existence that it's hard to philosophize it's hard to gain that distance and so we sometimes need for to wait for things to go wrong in order for as haiger says the structures of the world to illuminate to light up to become apparent and so when the Mouse runs out of battery at the worst possible B the controller drifts or lags and all of a sudden you know it's it's not just that you become aware of yourself as holding this object you kind of all of these different structures of the world light up you remember that this is a game you become more aware of the hit boxes uh the game physics all of the different little mechanics all of the numbers that the game constantly generating and it's like the whole world lights up and at the same time you have this kind of this tragic moment of distance and the interesting thing is that that breakdown is both an interruption but it's also super useful if you happen to be in a philosophical mood when the house the controller breaks down because then you can go aha now I'm aware of of all of these structures right no that's that's really convergent with yeah my my perspectives on this um yes I think the hammer example so he good gives this example of a hammer or hammering at Nails I also don't think it's the best example like the most canonical example of a handedness or or or definitely not flow I mean I guess Hyder is lacking examples because he didn't have probably didn't have rock climbers or Surfers or Gamers but I I'm hard pressed to imagine a hammerer so immersed in hammering I mean to be honest um I think it would involve a lot of yeah metacognitive processes what what we call epistemic foraging so a lot of exploratory Behavior even if that's mind wondering so I I I take your point I think the game example is a lot better and yes you're absolutely right it's not just that this uh rified sense of self emerges as you say the world itself shifts or or is modulated by that breakdown so I talk about how um the you know I give an example in my paper about a violin string breaking during a concerto and the violin had been in float you know the violin no longer seems to be presenting affordances so the violin itself seems to be

changed and suddenly you get all of this sort of concomitant self talk as well and you get all the Dows and you're suddenly very aware of the audience and the reason at least the reason why we uh postulate that that might be the case is because your free energy minimizing system has kind of entered into an epistemic mode it needs to resolve uncertainty about the world it doesn't have the availability of just like very high expected utility or or or pragmatic affordances so yes we can split this notion of affordances shout out of JJ Gibson into epistemic and pragmatic affordances and that's a really nice distinction with an active inference a question uh that popped into my head um actually the question before um I was I spoke to John BVI um a couple of weeks ago and that was published and it was a wonderful conversation and John was um really really insightful and I said something like well I think there's a uh intuition of over you know which guides us to see ourselves as some kind of cartesian ego and John and yes I think it's very much contextually bound but I think underlying that at least I think it's a you could propose that there is this intuition John said well I actually think that might be more of a cultural phenomenon um and actually it's a very Western idea and it's sort of built off the way that we structure society and the way that we see ourselves do you think that uh there's a kind of I won't use the word innate but instinctual or intuitive sense in which we model ourselves as a kind of rifed ego or do you think as John says it's very much bound up in context whether that be cultural or the kind of practice that we're undertaking at that moment yeah I think that's a fantastic so I don't know I I find myself kind of leaning in both directions right because on the one hand I do think there are significant cultural social historical factors that are going to shape this you know there's there's of course the you know that the kind of famous distinction of of um I guess Western if if you want to use that word individualism um as opposed to you know uh kind of other views of thinking about the self in relation to community um and uh I was I was reading a wonderful paper about Ubuntu Philosophy for example um which kind of you know says it's it's not it's not the cartesian thing of you know um I think therefore I am it's it's we are therefore I am right so that kind of fundamental community and relationality and so yes I I absolutely do think that there are um kind of specific factors influencing this on the other hand um so one of the interesting things that kind of I guess the phenomenological tradition which is also super eurocentric right um brings up is this idea that the way that cognition works kind of almost perversely encourages us to forget how cognition works right that it's it's this this kind of process these processes cover their own tracks and make us kind of you know Overlook just how practically immersed we are um and yeah there's this this kind of interesting idea of you know again for

example in in hiiga there's this idea that because we are fundamentally practical creatures that means we're likely to kind of lose ourselves in the Practical everyday to an extent that we start kind of you know forgetting the existential ramifications of what's happening and we we kind of start losing in a sense that kind of explicitly futural aspect of ourselves or you know and and we forget our mortality our finitude and all of that and you get this kind of throughout the phenomenological tradition that the very the very things the very um kind of purposiveness the embodiment the I guess the depth of our investment as well just how meaningful the world is to us meaning that kind of the precariousness of our condition and and all of these kind of the cogs underneath uh are things that we like to try and ignore and that we tend to turn away from and and that is also something that you know you again you do find in lots of philosophical traditions around the world this idea that we do kind of have this this maybe not innate but we certainly have a very common tendency to lose sight of ourselves forget ourselves in the very kinds of practices that you know are fundamental to the kinds of creatures that we are so there's always this kind of danger of losing sight of ourselves there's always this kind of danger of seeing ourselves in an atomized uh isolated uh individualistic way I was actually um just just earlier this year in in and ethics 101 we were going through um various uh forms of confusion ISM and there is this This brilliant um idea in in kind of uh one of the the the schools of confusion ISM that once you kind of start mistakenly thinking of yourself as an isolated object rather than as being part of the fabric of everything and being involved in in everything being relational then it be it it's almost this Snowball Effect where you actually become more and more self-isolating you feel lonelier because you're thinking of yourself right isolated that makes you more selfish and it has all of these kind of flowing and effects yeah so I guess yeah on the one hand I absolutely think there are immensely important historical social cultural specificities on the other hand I find it really interesting that lots of philosophical traditions talk about our tendency to kind of misconceive of ourselves these individual so that might be quite widespread yes that's that's that's really useful um yes I mean I guess it's it comes to a philosophical question about what selfhood is uh and how much we're being swayed by our own well it's only a of misconception or Illusion if you buy the notion in the first place that the self is a kind of atomistic rarified thing so again it's it's um it's kind of up for grabs but I think what's lovely about AC inference and again I'm going to toot its horn is that it kind of gives a computational account of this so I think what the way it would be answered now and I you know this is actually an area of work that I really want to keep delving into personally is that this hiigaran notion that we tend to forget exactly how cognition

is when we're immersed in the world I think that makes sense because what what we're doing when we do have a ray of f sense of self is accessing all the parts of our cognitive Machinery that rely on us on a memory on proposition right because we're in this epistemic mode we're resolving uncertainty we're testing out model parameters right these all these kind of technical things in active inference whereas when we're in flow or whether when we're in skillful coping to use dra's term we know we're not we're not deploying attention onto that part of the cognitive Machinery so it makes perfect sense why you know it's almost a contradiction a you know to say that um yeah to say that you could propositionally know who you are whilst you're immersed in the unfolding of reality that you're a part of so I think I think active inference actually gives a computational account of of where that of where that kind of Disconnect comes from but again I I I think it's I think the important thing to remember as well at least from an active inference perspective is that like one one is it really privileged from the other um because again one's a pragmatic mode and one's an epistemic mode and you need both to minimize free energy you need both to survive fundamentally um and so although yes that isolation that atomism is uh painful in a way uh and you know you could go all the way down to the Buddhism and say that the craving the dooka that comes from that self is the kind of Heart of suffering perhaps but at the same time uh I would rather be a creature that can balance epistemic and pragmatic needs than just I I I just don't think it's feasible given the type of things we are to be to be solely pragmatic cool well that's me that's my sort of rant over um yeah no I I I agree actually um yeah just sorry you jumped in just no no go ahead I just yeah I wanted to highlight um pretty much that that I agree and I think again that's kind kind of one of the interesting things especially about um you know heidig is account that kind of often gets missed there's not this idea that you know theoretical cognition is bad science is bad quantification is bad or that it's something that we could ever actually put aside um so again it kind of gets under intellectualized a lot and you know there's there's this tendency to kind of overemphasize like you know like you said kind of the importance of of uh immersed practical coping but the idea is but yeah we absolutely need both we need all of these states um and when things what happens when things go AR right philosophically speaking is when we forget that relationship and when we kind of focus on one to the exclusion of the other and forget how they interrelate so yeah so it's I I absolutely agree it's it's very much this sense of of yeah you you not only can we not put these Tendencies aside but we shouldn't want to we don't have to to they're part they're fundamentally part of what makes cognition as complex as it is right yeah yeah and again there's a kind of self-perpetuating loop perhaps in which when

we're in this kind of more epistemic mode we tend towards um wanting that certainty okay this is the way to be uh and I think we've seen this in the kind of westernization of something like a mindfulness practice what if I could just be mindful the whole you know whatever that means and it's a very L ter right depends on what mindfulness practice you're doing but also like will you being mindful to um how deep is that metacognition there are loads of questions that come up from that but my point is that like because we're seeking something to resolve our uncertainty and it's an existential uncertainty about how the world is again is a very decent thought in my mind we're looking for that kind of singular solution um because that's the way that's the way we resolve most of our uncertainty is just to find out the fact but actually maybe the deeper truth is that this these binary distinct like it's all it's all Dynamic you don't want to have one over the other and actually maybe that you know there's plenty of cases where I'm doing both pragmatic skillful coping and some epistemic exploration right when I open the door yes I'm not reifying the door handle I'm not doing epistemic foraging about the door handle but I might be about well I'm gonna have a dinner so occupy a very privileged position where there's this complete absorption and we have a kind of technical um detailing about why that might be the case I think the other thing is that um I don't know whether this is a strictly hiigaran idea I know draus talks about it is that this kind of dzign this being in the world brings a bit a sense of familiarity a uh you know draus would talk about skillful coping um Hyer talks about a-h handedness and as you say we're kind of yes we are practical creatures and the world is made for us and we're made for the world and there is this kind of mutual coupling so how fundamental is the notion of familiarity to phenomenology and I guess a question that comes up is are you know are we a risk of losing that familiarity and and if so how yeah um brilliant question thanks darus and I've I've actually been thinking about this too from kind of a different um direction of of of yeah kind of looking at different discourses that talk about the importance as active inference does too of course if if uncertainty and predictability and I think there's there's very much this need to again you know um as as we've been saying before about practice and Theory to resist binaries and then kind of resist you know prioritizing one uh over the other and so yeah on the one hand absolutely as you said you know um being the world involves in this this sense of not only that you know the world is made for us but more that you know we we're enacting this world this is this is our world that we're bringing forth and so of course you know the world that you never get is going to in a sense reflect yourself back at you it's your encountering it in terms of you know your needs your capacities your projects and so there is this this intimate familiarity with kind of practice

as such right with this idea of I mean that kind of um there's a point where hegger talks about in sense like that the weirdness if you really step back and think about it of the fact that yeah we're these creatures you just put us in the world and doing stuff in order to stay alive in order to striving for goals that's just something that we can naturally do that's just what it means to be alive and so yeah so there is this this kind of important sense of familiarity but at the same time um especially you know broadly in the phenomenological tradition um and you know kind of tangential theorist from husserl to again all the book on there there is this this emphasis on uncertainty as well so yes this is our world but we are fundamentally future directed creatures um we have to confront not just confront but Embrace uncertainty at every point and it's kind of that fundamental many ways that we're running away from a lot of the time but yeah there's this sense that you know and we we try to reduce our relationship to uncertainty and unpredictability by just cashing it out as planning anticipation I can you know I can tame the week by writing down exactly what's going to happen but of course life doesn't work like that so you know hegger talks about confronting and and kind of being directed to possibility as such um and this importance of these moments where the world isn't familiar anymore right so he he talks about UNH um which you know is is kind of it's it's it's horror it's eeriness um but it it is quite literally not at homeness right so where that familiarity that we take for granted is kind of the rug is pulled out from under us and we need those moments again not only to do philosophy but actually to get a sense of the existential import of what we're confronting so there is this kind of constant toity of familiarity yes we're intimately familiar with practice as practical beings but also with creatures of uncertainty we're creatures of unpredictability and that is kind of what it means to be alive to confront and and grapple with unpredictability all the time well I'm so happy you brought up the future as well there because I think uh that's that's kind of one of the big themes I do want to touch upon before I do that um and so that was kind of just a memo to myself I think a lot of what you're talking about here in terms of Heidegger maps onto his account of mood um am I I think I might be speaking speaking out of place it but I think H puts a real emphasis on anxiety and boredom almost like these are these kind of two axes uh or or the end of the axis of mood is that right and if so why those two moods uh seems one quite pessimistic um and two maybe not totally faithful to our lived experience where there's joy and relief and other experiences yeah no that's a that's a real question yeah that's um that's exactly right so that's the the the kind of part of his account his framework that I'm looking at here right this this idea of you know the the world matters to us and it matters to us in a particular way because we're attuned in a particular way and yeah

there are some moods that kind of particularly fundamental uh especially from the perspective of somebody who wants to make the everyday just strange enough to be able to do philosophy about it and so Heidegger does talk about you know other effective states he does talk about joy and love and anger and hate and all of that sort of stuff but the existential import of anxiety and boredom I think is is yeah the capacity to to kind of again be like the broken Hammer right cast the world into light by kind of by disrupting how we navigate kind of everyday sense making in particularly radical ways right and so especially with anxiety you know the the key is that you're you're not afraid for yourself as such you're kind of you're experiencing this dread at the loss of or at you know at the at your world at your normal ways of sense making being put into question um which reveals how those uh operations of of sense making actually function um and so yeah so it's it seems like a particularly PR account but um again that's it's something you know that you find in in in phenomenology I mean um what was it I'm trying to remember the three ones that SRA focused on in particular so there were shame um as you know a shame being kind of the proof of the existence of other Minds right so we could only feel shame if other Minds actually stood because nothing could make us feel shame the way that another free like human being another for itself could and so it's it's that sense of you know it's not deliberately it's not saying this is all there is to The Human Condition it's not saying these are the only important effective States but it's saying these are particularly significant because of what they show us about how the world and relational cognition operate yes so they were cheery weren't they um John Paul thank you for that well yes you did say uh you must act as if the whole world is watching you which is slightly slightly ominous um maybe true slightly ominous though um great well yes let's jump to Futurity uh my word of the week because I've been reading your papers um you quote Hyer who says that self- projection upon for the sake of itself again back to the hyphens is grounded in the future and is an essential characteristic of existentiality the primary meaning of existentiality is the future you then also say and I'll get it up secondly I think that we can also read the temporality of autopoietic cognition in future weighted terms the meaningful World generated by the activity of the organism in collaboration with its environment is arguably defined by an orientation towards Futurity in that the future survivor of the system that provides the framework for all its actions and the salience that it produces in conjunction with the situations it encounters so we got kind of two strands there and a dual Focus on the future one from Heidegger and one from the inactivist um tradition obviously active inference is the future and the unfolding the future is very important in active but before we go there um let's

start with haiger why is the future such a critical part of his philosophy yeah uh great question darus so haig's account uh of time in general just to to kind of step back a bit um is an account that fundamentally fundamentally looks at live time in kind of nonlinear terms right so the idea is that um you can't kind of nicely neatly separate past present and future into these uh you know these these kind of separable moments they're they always intertwined and they have these feedback loops in a sense where you know past and future kind of shape each other in in ways that we wouldn't expect if we had a purely linear conception of time and and just just to you know to to kind of to add a little S note to that I think so just from the little that I've read of Act of inference that seems again a particularly interesting point of resonance right this idea of you know past and future kind of revising each other because what you're what you experience what you're acting what you're doing your predictions kind of help you to revise your priz and there this kind of yeah feedback loop of like updating both predictions and and and kind of Prior beliefs just to jump in there before I forget um I think a paper that actually you would really enjoy but having everyone would really enjoy is um yakob how's got one called distrusting the present where he basically takes the phenomenology of um I think William James called it the specious present and I know H had a you know hle talked about this kind of seeping of the past and the future into the present and how this yields a kind of flow of time and he gives a theoretical account of that under active inference terms in terms of attention and precision waiting so just little addendum there for anyone who's interested um active inference has dabbled into the phenomenology of time and that's a really good place to start but anyway back to back to hiiga yeah cool no that's that's super interesting I'll have to I'll have to check out that send we we're kind of we're operating with this you know nonlinear sense of of time where all of these different aspects of of time can revise each other so kind of moving away from this you know focus on the now and this this focus on need linear sequences um and within that though so on the one hand haiger argues for what he calls in typically hiigaran terms the equ primordiality of past present and future which just means they're you know equally primordial you you're not just focused on the now each of these aspects is kind of meant to be um yeah equally significant equally fundamental in in kind of structuring the way that we experience things um but then there is within all of that still this this kind of emphasis on the future the stretching into the future and um and you know so how assess that explicitly a whole bit but then of course the question is why does he say that explicitly a whole bit and so um part of the the reason for that I argue is is that the conception of ourselves or of you know of the living organism I don't think that this is just restricted to the

human the conception of the living organism as a creature that has that fundamentally wants to stay alive that's concerned about itself that has projects to fulfill whether that project is you know going from here to there or you know I want to become a bestselling writer or whatever someone's existential project might be the very idea of a project is is always future directed so if we're defined the very idea of self-concern the very idea of persistence uh self-maintenance survival is always kind of going to be fundamentally future it doesn't make sense except in futural terms and the reason that we need to try and survive the reason that persistence is an effort the reason that Annihilation is a Potential Threat is because of uncertainty right if there were no openness uh if there were no sense of uh something that could happen uh that you need to adjust your responsiveness to then there would you know then there would just be stagnation there would be nothing um and and so and again going back to that concept of adaptivity that we've touched on a few times um and this is why I think this is particularly resonant there there are particular resonances between hiigaran phenomenology and an activism this very idea of being able to respond to uh what happens and of course for hiig all of that is caught up in the overarching importance of our limitations our precariousness our finitude our mortality right where we be because there is this ultimate end point to our future and everything before that is open-ended that's kind of the the reason the ultimate kind of structural orientation that shapes everything else that falls into place around it so yeah you've you've got like I said on the one hand this nonlinear account of time but everything about it privileges the future because it's the future in a sense kind of explains cognition explains life you couldn't be the kind of creature that we are you couldn't be a living creature without this kind of future directedness and and having to Grapple with uncertainty great um and I guess before we make that convergent point with AO pris activism which which should be becoming clearer for the audience I think what's really uh special about this account that H H gives us is and I think it's your term which I completely love is an ongoing Futurity I said that to my master supervisor um because I have only done my master sorry you know haven't done my PhD yet um so yes still very much a layment to all of these but I said it to my master supervisor because he's uh he's loves Hy and actually did his doctoral thesis on skillful coping um and situated cognition so this is this would really be up his street I said I I absolutely love this I think this is an amazing idea because yes you're right I mean we think of maybe again it's this cognitive bias the past the present the future as these kind of uh I don't know these kind of rarified extensions and and the future kind of going on ad infinitum but haig's got this different perspective of an ongoing Futurity and I think if I read your work correctly it points

to um it points to this notion that everything we have to do has to be grounded in some sense because we're we're fundamentally self-concerned and to be self-c concerned is to um take into account the future of yourself right there's no self there's no self-c concern without a reference point for what you're going to be at time n plus one so maybe it would be really uh useful because I probably butchered that to just give a little bit more about what this concept is of kind of an ongoing Futurity awesome thank you and you didn't butcher that at all that was actually really incredibly well articulated no that was awesome thank you so I I don't have much to to add on on top of what you've said but yeah just to to kind of essentially to to highlight what you've mentioned exactly as you said the very idea of of self-concern means that there's always going to be this this future directedness you know there's like the all of these Concepts that we talk about in you know embodied cognition and activism active inference to the you know all of them early Mak sense in terms of a relationship between n and n plus one right or T and $t+$ one the current state of the system and the next state of the system is that a desirable relation uh is it an undesirable relation what does the system do to get from this state to that state what happens when that goes wrong motivation normativity agency action all of those only make sense in in in those terms they're all about relations between the current and futural states of of the organism um and exactly as I said for for haiger the future and self-c concern are fundamentally Linked In These ways and there is very much this emphasis that this is this is a a Futurity that stretches out until it doesn't right so it's never a future that arrives and and I think that especially that's something that appeals to me about an activism this idea that stagnation is death right that there is the system's not going to stop changing um it can't the system is always going to be a responding to stimuli it's always going to be grappling with uncertainty um and I guess that's also something that happens in active inference you're always you know the brain doesn't suddenly go okay I've I've predicted it I've done it I'm done box is ticked I'm going to sit here now and that's it um that only happens in that only happens when the system ends yeah um that yeah it's it's always it's always this this kind of this year this this stretched out Futurity and even when you accomplish a particular goal that you know that still is kind of organized in this broader framework of future directedness of goal directedness where there's always this overarching goal of of self-c concern that's not something you can achieve that's not something you can be done with right you're only done with it when it ends yeah yeah yeah exactly that's wonderfully uh well said yeah our being morbid it does sound nice to just have a bit of stasis though but unfortunately no unfortunately we're just open and we're going to keep being openly Dynamic um I think

just to you know cap off what we've been talking about here in terms of this ongoing Futurity if I may um I'd like to read the sort of extension of that b you know I talked about autoparesis being in couched in future rated terms you say um the survival of the organism is not merely a simple not yet that is to be anticipated or drawn out in a concrete plan however this is a future that will never arrive or be finally accomplished it is an ongoing Futurity that orients everything else and that can never be closed off for the striving towards it can only end not with being grasped or secured for these are not possible but with the annihilation of the framework they have rendered meaningful with the disintegration of the system for the autopoietic Adaptive system stasis is death there is O there is no end point only the ultimate ceasing of those processes of resistance stance and precariousness that Define the very the organism's very life ah I love that that's that's so good um oh thank you no it's it's really beautiful what I will say uh again I'm not making a criticism but I think it's just interesting is that you also say haig's account again we'll go jump in you know without viewing Hyer the activ kind of one extension but jumping back to Hyer Hyer account advis us to view the futureal dimension as having a special significance one that be can be cashed out in terms of a radical indeterminacy that transcends mere predictive or anticipatory models of Futurity now I'm sure everyone's ears are just going mental because they heard the word predictive and we will get very excited when we hear predictive so yes I take the philosophical and phenomenological point that there is uh yes that being itself has to contend with the indeterminacy of the future and and that irregularity that unpredictability shapes itself concerned in the sense that everything is is uh projected to some $t+1$ I kind of I I I sense that there isn't actually a conflict here or attention with active inference um because I think we're probably talking about two different things here one is that we have the kind of uh very mechanistic technical uh capability to predict in some t uh deep temporal way and then the other is actually what is the phenomenological experience in terms of like how I live or my living is uh Guided by some future orientated self-organization is that would you say that's a kind of sufficient way to to resolve that not under playing of uh prediction but let's just say like that prediction itself is not sufficient for a hiigaran account of uh being yeah I I think that's a great way of putting it so especially yeah that that last Point as you said so that prediction is not sufficient um so it is it is worth noting that you know when um when heidig for example uses words like anticipation or prediction he's trying to contrast different ways that we have of relating uncertainty right and so um yeah where the the kind of the the more flatter or more reductive ways of relating to uncertainty or in a sense about trying to tame uncertainty um and

so if there's I assume that you know that the way that we'd be using prediction that circumstance isn't the way that you know prediction is cashed out in for example predictive processing that it it's I'm assuming that you know that is a much kind of richer more complicated process as well especially where things can go wrong right so the the possibility for yeah for for things to go arrive for what happens to be other than predicted for there to be risk there to be failure that's particularly important and um yeah for a relationship to again to what it is that we're predicting that is you know is is less about kind of plotting out a straightforward path that rules Out Future uncertainty and is kind of more I don't want to use the word surfing uncertainty because that immediately sounds like the the title of Andy clar's wonderful book um but yeah that's that's not what I mean you know embracing uncertainty yeah yeah without rolling out future uncertainty and so I guess yeah if there's room in prediction as it's cashed out in in active inference for more richer ways of relating to the Future then yeah there's not really a conflict here no exactly yeah I I think the hiigaran um anxiety of prediction is probably very cable within a broader mathematical understanding of what prediction or prediction error is in in active influence I think it's just good to iron that out um in case anyone uh stumbles across that the final thing you know I know it's getting late you're in Australia I'm in London it's not getting late here the sun's coming up um is much more a kind of meta conversation about philosophy of science um this is actually really I you know really what your doctoral thesis and later works have been looking at which is what is the possibility for um finding convergence points or finding enriching dialogue between phenomenology and Natural Sciences and one uh approach has been to say that the Natural Sciences are explanatory and phenomenology is descriptive um I know that's not the approach that you or other contemporary phenomenologists adopt if uh given that why why is that a problematic uh Duality excellent I'm so glad that you bring this up and so yes so that's uh something I've been turning my attention to more recently so I've got a paper specifically about phenomenologically motivated views of of explanation and so I think um ultimately that distinction of of kind of relegating phenomenology to whatever is not explanatory ends up missing out on a lot of interesting resources that both the phenomenological tradition and contemporary phenomenology have um in it's it ends up kind of giving us this reductive view of phenomenology where you know phenomenology is often cashed out as kind of just describing what a particular experience is like um and you know which uh and something I point out in in this paper is which is is dangerous not just because that's not what phenomenology does phenomenology is a lot more rigorous and systematic it's about looking

at what's underneath those experiences is what's not obvious The Experience um but you know if if we start talking about so Dan Harvey makes this this great point that phenomenology is not just about the phenomenon phenomenal and if we just start restricting phenomenology to describing the phenomenal we might make all sorts of commitments to what that phenomenal is we might you know think phenomenology is just about qualia um a concept that I'm always very wary of um and and not as interested in as as as um uh as maybe I'm I'm expected to be um but also generally I think there are models and instances of phenomenological explanations that are genuinely useful insightful um about the mechanisms that explain why phenomena are the way that they are and I think that's especially the case if we kind of complicate how we think of explanations and that's where where um I argue that phenomenology and and an activism again kind of converge right and and I assume again that that active inference has you know lots of useful things to say here too because it's about looking away from that very linear asymmetrical model of explanation of here is the cause here is the effect they're kind of you know a nice neat sequence in time they're linear they're cut off from each other right it's what all of these discourses about cognition show us whether it's in activism whether it's active inference is as we've already talked about there's a lot more complex feedback between past and present and future there's a lot more feedback between cause and effect um and so yeah we need models of explanation that are able to trace out those interesting feedback loops and so I think phenomenology especially has a lot to offer in in kind of that regard yeah I guess um someone listening might be curious about if there's a kind of example in the literature where phenomenology has been able to give an explanatory account of a phenomenon that might otherwise be sort of only in the purview of the Natural Sciences just wondering if if if that you know if if you can think of one off the top of your head yeah so in uh in in that paper that I mentioned I look at various accounts psychological explanation um they also look at a couple of of recent attempts to um conceptualize Evolution or kind of to take up an inactive uh concept of evolution and to look at things like minimal and basal cognition um but perhaps for a kind of more accessible example I think just going back to the history of the development of embodied cognition is is always is always useful and this is kind of you know something that comes up whenever or that I try to throw in whenever people talk about the dialogue between phenomenology and cognitive science or the Natural Sciences in general what could phenomenology actually offer um what could it bring that might be missed otherwise and I mean the you know embodied cognition developed the way that it did because um people read phenomenology and you know because Dyers and you know the the

the mid 20th century thought hey look this this's stuff that these problems we're running into when we're trying to develop you know the most kind of primitive forms of of of artificial systems um they're actually problems that I kind of predicted and and addressed and and set out in all of these different phenomenological Traditions from from Heidegger to to Merleau-Ponty and you know of course um I mean for for Heidegger that kind of pointed to an almost irresolvable problem um but yeah the you know embodied cognition has taken the shape that it has as a kind of huge set of different discourses because these these models of the relation between cognizer and body and world as they were cashed out and phenomenology proved useful so that I think that in itself is kind of a a useful like metadiscursive example yeah yeah I mean draw when do what computers can't do yet as like the early 70s yeah that's right I love so it's still still my my my favorite sequel um idea for a SQL title right so what computers can't do followed by what computers still can't do yeah what computers still can't do I mean yeah it's fascinating again it's a complete you know it could be another hour and a half just on that um but you're right I mean the the as far as I can tell the AI folk are struggling um with Just LLMs For Stuff where you might want a little bit more embodiment and you can um you know we can point people to a company like Versus who for example are applying active inference uh the active inference framework to artificial intelligence and hopefully there are going to be some you know I'm sure exciting developments already happening but I think the future is bright um for the role of embodied cognition in developing um intelligent agents intelligent artifacts Mar this was uh this was super I mean for me like it's an opportunity to discuss haiger with a hiigaran is wonderful again I uh I'm a complete Layman um I give a lot of credit to my master supervisor again I only you know only finished my Masters this year but actually um you know I can imagine having just left University to have an educator like you would be a complete privilege so thank you so much I actually learned a wonderful amount um and I mean actually before you go maybe let people know you I said you you're lecturing philosophy one on one will you focusing on the moment what can people keep their eyes out for in terms of your output awesome thank you so much so first thing I wanted to thank you for having me on this has been absolutely fantastic it's been such a fun conversation and you're so brilliant to talk to than I think it's so cool that you're doing this at this stage in in your career I thought you were you know a post talk or something so you certainly don't come across as a Layman and this is like a really cool initiative um and so yeah so in terms of what people can keep an eye out for um so I do have a book on Heidegger in time coming out co-written with my wonderful friend Emily Hughes uh it's coming at April next year through Routledge so hiig is alternative

history of time I should plug um and yeah and other than that I'm working on a whole bunch of projects actually looking at synesthesia and counteracts among other things so yeah nice all right well that that book is extremely exciting and thank you so much for your kind words yeah I mean um all credit goes to the The Institute um for letting me do this yeah I mean to be honest uh you know I've never actually announced who I am we sort of just kicked off this series with Carl as me being complete random yeah I mean I yeah as I said earlier I mean I finished my masters in September but discovered active influence about a year ago and has have just absolutely fall in love with it and just its capacity to bring into the picture insights from phenomenology philosophy physics maths which I think is exactly the interdisciplinary approach that people have been talking about but haven't been fulfilling their promise on um but thank you that's that's really very kind of you and and yeah I mean hopefully uh we're going to keep getting wonderful guests Like You on to to educate me and our audience even more uh Marilyn thank you so much once again thank you so much darus it's been awesome cool thank you guys thank you